unit - III

Cognitive Modeling:

Modeling the Interaction of Language:

⇒ It refers to designing computational systems that simulate how humans use, understand, & respond to language in interaction (like conversations).

- It focuses on how people:
  - understand language in content.
  - Responds appropriately.
  - Maintains conversations.

⟹ The Language interaction means communication b/w two (or) more participants, such as:
- Human ⟷ Human
- Human ⟷ Machine.

- It involves:
  - understanding meaning
  - managing content.
  - Taking turns in conversation.

Cognitive Modeling Approach:

→ cognitive Models try to replicate human conversational behaviour using computational techniques.

- It stimulates how the brain:
  - processes I/P
  - Interprets meaning
  - Generates responses.

computational techniques used:

(i) Rule Based Models:
- In this there will be predefined conversation rules.
- It is used in simple chatbots.

(ii) Probabilistics Models :-

- In this model, it predicts responses based on probability.

  ex: N-grams.

(iii) Neural N/w models :-

- ~~DL~~ These are DL based models.
- It Learn's from large ~~data~~ language datasets

(iv) Reinforcement Learning :-

- It learns best responses through feedback.

Components of Language Interaction Modeling :-

(i) Perception (I/P processing) :-

- Understanding spoken (or) written language.
- It uses NLP techniques like Parsing & speech recognition.

(ii) Comprehension :-

- Extracting meaning from Sentences.
  - It uses : • syntax (structure)
    • semantic (meaning)
    • Paragmatic (content).

(iii) Content Management :-

- Keeps track of previous conversation.
- Helps in understanding references.

Ex: "Where is my book?" → "It is on the table".
("It" refers to book)

(iv) Dialogue mangement :

- controls the flow of conversation.
- Decides :- what to say next
  - what to respond.

(v) Response Generation :-

- Produces meaninful replies.
- Uses :- Rule based systems.
  - Neural language models.

Ex: conversation :- user : "what is the weather today"?
system : 1) understands qtn
2) Retrieves weather data.
3) Responds : "It is sunny today!"

Applications :

- chatbots
- virtual Assistants
- Machine translation.

Importance :

- It enables natural communication with machines
- Improves user experence.
- Makes AI sys intelligent.

Memory & Learning :-

⇒ In cognitive modeling memory & Learning are fundamental concepts that help machines simulate how humans store info & improve performace through experience.

Memory in cognitive Modeling :-

→ Memory is the ability ~~to store~~ of a sys to store, retain & recall info when needed.

Types :-

(i) Short term memory (STM) :-

- It stores temporary information.
- It has Limited capacity.
- It is used during reasoning (or) problem solving.

ex:- remembering a number for a few seconds.

(ii) Long term memory :-

- Stores information for a long time & its divided into :-

(a) Episodic memory :-

- Stores personal experiences / events.
- Includes time & content.

(b) Semantic Memory:
- Stores facts & GK.

(c) procedural memory:-
- Stores skills & actions
ex: riding a bike

Computational Representation:
- Memory in AI sys is modeled using:-

- DB & knowledge bases
- Neural N/w weights.
- memory N/w. (RNN, LSTM)
- knowledge graphs.

Learning in cognitive Modeling:
- It is the process of improving performance based on experience (or) data.

Types:
(1) Supervised Learning:-
- Learns from labeled data.
ex: classification tasks.

(2) Unsupervised Learning:- Learns from unlabeled data
ex: clustering.

(3) Reinforcement Learning:- Learns through rewards & punishments.
ex: game playing AI.

(4) Cognitive learning:- Mimic human Learning.
- uses reasoning & understanding.

Role in CCL

- They enables sys to adapt & improve.
- Supports decision making.
- Helps in content awareness
- Makes AI sys more human like

Applications:

- Chatbots & virtual assistants
- Recommendation sys
- Speech & image recognition.

Modeling select aspects of cognition: classical models of rationality:

⇒ In CC, modeling cognition means representing how humans think & make decisions.

- one imp approach is classical models of rationality, which assumes that humans make logical & optimal decisions based on available info.

Rationality: - It means:
- making decisions using logic & reasoning
- Choosing the best possible option to acheiv a goal.

Scanned with OKEN Scanner

⇒ In classical models, humans are seen as a perfect decision makers.

Classical models of Rationality :

(i) logic based models :

- It uses formal logic to represent thinking.
- Decisions are made using rules.

ex: "If 'A' is true. B is true ⟶ then 'C' is true".

- It is used in expert sys. & Rule based AI.

(ii) Bayesian Decision Models :

- It uses probability to handle uncertainity.
- It updates belief's when new info is available.
- It is based on Bayes' rule.
- It is widely used in AI reasoning.

(iii) optimization Models :

- Finds best solution among many options.
- It uses mathematical techniques.

ex: shortest path, cost minimization.

Limitations :

⇒ Humans donot always behave rationally.

- Real world decisions involves : - emotions.
- This led to bounded rationality models (more realistic).

Role in CC :

- It forms the foundations of early AI sys.
- used in : - expert sys
  - planning Algo.

Applications :

- Medical diagonosis sys
- Financial analysis.
- Business decision Making systems.

Symbolic reasoning & Decision Making :

⇒ In CC, Symbolic reasioning referes to represlenting knowledge using ~~sysmb~~ symbols & applying reasoning to make decisions.

- It is one of the classical AI Approach where sys think in a human - like logical way.

⇒ "Symbolic reasoning uses :
- symbols (like words, obj, variables)
- Rules & logic

to represent knowledge & perform reasoning.

ex : "If it is raining → "take an umberalla".

⇒ In this knowledge is stored in symbolic forms

- Reasoning is done using logical rules.

Components of symbolic Reasoning :-

(i) Knowledge representation :- - Information stored as :-

- facts
- Rules & logical stmts.

(ii) Inference mechanisms :-

- Applies rules to derive new knowledge.

Types :- • forward chaining (data → conclusion)

• Backward " (goal → data).

(iii) Logic systems :-

- use formal logic like :- • prepositional logic
• Predicate "

Decision Making in symbolic sys :-

- Decision making is the process of choosing the best action based on reasoning.

Working :- ① Represent problem using symbols
② Apply logical rules
③ Infer possible outcomes
④ Choose the best solution

Ex: Medical diagnosis sys:-

- symptoms → rules → disease identification.

Types of Decision Making:-

(i) Rule based DM:-

- It uses predefined rules.

(ii) logical based DM:-

- uses logical inference

(iii) Goal Based DM:-

- Chooses actions to achieve a goal.

Advantages:-

- Easy to understand & interpret.
- logical & structured.

Limitations:-

- can't handle uncertainity well.
- Requires predefined knowledge.
- Not suitable for complex data.

Applications:-

- Expert sys
- Chatbots
- Medical diagnosis
- Game playing systems